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================================================================================================================================================
  CONVEX-ROOF C_GME  -- METHOD-2 (Exponential Map)
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  Generated          : 2026-04-10  20:59
  System             : 1D Hubbard ring  (mode: uniform)

  SYSTEM PARAMETERS
  ------------------------------------------------
  Sites          L   = 5
  Electrons     Ne   = 5
  Hopping        t   = 1.0
  Interaction    U   = 4.0

  GROUND STATE
  ------------------------------------------------
  E0                 = -2.6000813248
  E1                 = -2.6000813248
  Gap  E1-E0         = 0.0000000000
  <S^2>              = 0.750000
  S                  = 0.5000
  <Sz>               = 0.250000
  Multiplicity       = 2.0
  Spin state         = Doublet (S=1/2)
  <D> total          = 0.459102
  <D> / site         = 0.091820

  ANALYSIS PARAMETERS
  ------------------------------------------------
  Clusters analysed  = 5
  Optimization Methods used       = 6
  Restarts / method  = 20

  OPTIMIZATION METHODS
  ------------------------------------------------
  1. L-BFGS-B
  2. Nelder-Mead
  3. Powell
  4. COBYLA
  5. Differential-Evo
  6. Simulated-Ann

  NOTATION
  ------------------------------------------------
  C_LB     = Lower bound  (paper coherence-witness, analytical)
  C_GME    = GME for pure rdm
  C_UB     = Upper bound  (eigendecomposition, analytical)
  C_GME_X  = Convex-roof estimate from optimiser X

  True C_GME satisfies:  C_LB <= C_GME <= C_UB

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CLUSTER OVERVIEW -- All Optimizers

Cluster	Purity	C_LB	C_GME	C_UB	C_GME_L-BFGS-B	C_GME_Nelder-M	C_GME_Powell	C_GME_COBYLA	C_GME_Diff-Evo	C_GME_Sim-Ann	BestMethod	Status
(0, 1, 2, 3, 4)	1.00000000	0.00000000	0.49633318	0.49633318	* 0.49633318	0.49633318	0.49633318	0.49633318	0.49633318	0.49633318	L-BFGS-B	MIX
(1, 2, 3, 4, 0)	1.00000000	0.00000000	0.49633318	0.49633318	* 0.49633318	0.49633318	0.49633318	0.49633318	0.49633318	0.49633318	L-BFGS-B	MIX
(2, 3, 4, 0, 1)	1.00000000	0.00000000	0.49633318	0.49633318	* 0.49633318	0.49633318	0.49633318	0.49633318	0.49633318	0.49633318	L-BFGS-B	MIX
(3, 4, 0, 1, 2)	1.00000000	0.00000000	0.49633318	0.49633318	* 0.49633318	0.49633318	0.49633318	0.49633318	0.49633318	0.49633318	L-BFGS-B	MIX
(4, 0, 1, 2, 3)	1.00000000	0.00000000	0.49633318	0.49633318	* 0.49633318	0.49633318	0.49633318	0.49633318	0.49633318	0.49633318	L-BFGS-B	MIX

* = best (lowest) C_GME for that cluster

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  Optimizers SUMMARY  (aggregated across all clusters)
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  Optimizer            Mean C_GME    Std Dev         Best        Worst   Total(s)  Rank
  ------------------------------------------------------------------------------------
  L-BFGS-B             0.49633318   0.000000   0.49633318   0.49633318       0.00     1
  Nelder-Mead          0.49633318   0.000000   0.49633318   0.49633318       0.00     2
  Powell               0.49633318   0.000000   0.49633318   0.49633318       0.00     3
  COBYLA               0.49633318   0.000000   0.49633318   0.49633318       0.00     4
  Differential-Evo     0.49633318   0.000000   0.49633318   0.49633318       0.00     5
  Simulated-Ann        0.49633318   0.000000   0.49633318   0.49633318       0.00     6

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  TIMING ANALYSIS
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  Optimizer            Total(s)   Avg/cluster(s)  Fraction(%)  Rank(quality)
  ------------------------------------------------------------------------------
  L-BFGS-B                 0.00             0.00          0.0              1
  Nelder-Mead              0.00             0.00          0.0              2
  Powell                   0.00             0.00          0.0              3
  COBYLA                   0.00             0.00          0.0              4
  Differential-Evo         0.00             0.00          0.0              5
  Simulated-Ann            0.00             0.00          0.0              6
  ------------------------------------------------------------------------------
  TOTAL                    0.00


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================================================================================================================================================
  CLUSTER DETAIL  --  Sites (0, 1, 2, 3, 4)
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  n (body order) = 5
  Purity Tr(r^2) = 1.0000000000
  Rank           = 1
  C_LB (lower)   = 0.0000000000
  C_GME   = 0.4963331775
  C_UB (upper)   = 0.4963331775
  Best C_GME (optimizer)     = 0.4963331775  [L-BFGS-B]
  Bounds : C_LB <= C_GME <= C_UB?     = YES

  Bipartition purities
  ------------------------------------------------------------
  A                    B                       Tr(rho_A^2)
  ------------------------------------------------------------
  (0,)                 (1, 2, 3, 4)            0.6210991407
  (1,)                 (0, 2, 3, 4)            0.5445861398
  (2,)                 (0, 1, 3, 4)            0.5445861398
  (3,)                 (0, 1, 2, 4)            0.5445861398
  (4,)                 (0, 1, 2, 3)            0.5445861398
  (0, 1)               (2, 3, 4)               0.7757053517
  (0, 2)               (1, 3, 4)               0.4859408796
  (0, 3)               (1, 2, 4)               0.4859408796
  (0, 4)               (1, 2, 3)               0.4859408796
  (1, 2)               (0, 3, 4)               0.4859408796
  (1, 3)               (0, 2, 4)               0.4859408796
  (1, 4)               (0, 2, 3)               0.4859408796
  (2, 3)               (0, 1, 4)               0.4335780367
  (2, 4)               (0, 1, 3)               0.4335780367
  (3, 4)               (0, 1, 2)               0.4335780367

  Method results
  ------------------------------------------------------------------------------------
  Method                      C_GME   Time(s)  Restarts   Evals       Spread
  ------------------------------------------------------------------------------
  L-BFGS-B             0.4963331775      0.00         0       0     0.00e+00
  Nelder-Mead          0.4963331775      0.00         0       0     0.00e+00
  Powell               0.4963331775      0.00         0       0     0.00e+00
  COBYLA               0.4963331775      0.00         0       0     0.00e+00
  Differential-Evo     0.4963331775      0.00         0       0     0.00e+00
  Simulated-Ann        0.4963331775      0.00         0       0     0.00e+00

  Pure-state decomposition  [L-BFGS-B]
  ------------------------------------------------------------
     #     weight p_i   C_GME(|psi_i>)      p_i*C_GME
  ------------------------------------------------------
     1   1.0000000000     0.4963331775   0.4963331775
  ------------------------------------------------------
   Sum                                  0.4963331775
  ( best C_GME = 0.4963331775)

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================================================================================================================================================
  CLUSTER DETAIL  --  Sites (1, 2, 3, 4, 0)
================================================================================================================================================

  n (body order) = 5
  Purity Tr(r^2) = 1.0000000000
  Rank           = 1
  C_LB (lower)   = 0.0000000000
  C_GME   = 0.4963331775
  C_UB (upper)   = 0.4963331775
  Best C_GME (optimizer)     = 0.4963331775  [L-BFGS-B]
  Bounds : C_LB <= C_GME <= C_UB?     = YES

  Bipartition purities
  ------------------------------------------------------------
  A                    B                      Tr(rho_A^2)
  ------------------------------------------------------------
  (0,)                 (1, 2, 3, 4)           0.5255223136
  (1,)                 (0, 2, 3, 4)           0.6210991407
  (2,)                 (0, 1, 3, 4)           0.6210991407
  (3,)                 (0, 1, 2, 4)           0.6210991407
  (4,)                 (0, 1, 2, 3)           0.6210991407
  (0, 1)               (2, 3, 4)              0.3283275934
  (0, 2)               (1, 3, 4)              0.2851006472
  (0, 3)               (1, 2, 4)              0.2851006472
  (0, 4)               (1, 2, 3)              0.2851006472
  (1, 2)               (0, 3, 4)              0.2851006472
  (1, 3)               (0, 2, 4)              0.2851006472
  (1, 4)               (0, 2, 3)              0.2851006472
  (2, 3)               (0, 1, 4)              0.5707360127
  (2, 4)               (0, 1, 3)              0.5707360127
  (3, 4)               (0, 1, 2)              0.5707360127

  Method results
  ------------------------------------------------------------------------------------
  Method                      C_GME   Time(s)  Restarts   Evals       Spread
  ------------------------------------------------------------------------------
  L-BFGS-B             0.4963331775      0.00         0       0     0.00e+00
  Nelder-Mead          0.4963331775      0.00         0       0     0.00e+00
  Powell               0.4963331775      0.00         0       0     0.00e+00
  COBYLA               0.4963331775      0.00         0       0     0.00e+00
  Differential-Evo     0.4963331775      0.00         0       0     0.00e+00
  Simulated-Ann        0.4963331775      0.00         0       0     0.00e+00

  Pure-state decomposition  [L-BFGS-B]
  ------------------------------------------------------------
     #     weight p_i   C_GME(|psi_i>)      p_i*C_GME
  ------------------------------------------------------
     1   1.0000000000     0.4963331775   0.4963331775
  ------------------------------------------------------
   Sum                                  0.4963331775
  ( best C_GME = 0.4963331775)

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================================================================================================================================================
  CLUSTER DETAIL  --  Sites (2, 3, 4, 0, 1)
================================================================================================================================================

  n (body order) = 5
  Purity Tr(r^2) = 1.0000000000
  Rank           = 1
  C_LB (lower)   = 0.0000000000
  C_GME   = 0.4963331775
  C_UB (upper)   = 0.4963331775
  Best C_GME (optimizer)     = 0.4963331775  [L-BFGS-B]
  Bounds : C_LB <= C_GME <= C_UB?     = YES

  Bipartition purities
  ------------------------------------------------------------
  A                    B                       Tr(rho_A^2)
  ------------------------------------------------------------
  (0,)                 (1, 2, 3, 4)            0.5089904672
  (1,)                 (0, 2, 3, 4)            0.5255223136
  (2,)                 (0, 1, 3, 4)            0.5255223136
  (3,)                 (0, 1, 2, 4)            0.5255223136
  (4,)                 (0, 1, 2, 3)            0.5255223136
  (0, 1)               (2, 3, 4)               0.8768266884
  (0, 2)               (1, 3, 4)               0.3211372865
  (0, 3)               (1, 2, 4)               0.3211372865
  (0, 4)               (1, 2, 3)               0.3211372865
  (1, 2)               (0, 3, 4)               0.3211372865
  (1, 3)               (0, 2, 4)               0.3211372865
  (1, 4)               (0, 2, 3)               0.3211372865
  (2, 3)               (0, 1, 4)               0.7757053517
  (2, 4)               (0, 1, 3)               0.7757053517
  (3, 4)               (0, 1, 2)               0.7757053517

  Method results
  ------------------------------------------------------------------------------------
  Method                      C_GME   Time(s)  Restarts   Evals       Spread
  ------------------------------------------------------------------------------
  L-BFGS-B             0.4963331775      0.00         0       0     0.00e+00
  Nelder-Mead          0.4963331775      0.00         0       0     0.00e+00
  Powell               0.4963331775      0.00         0       0     0.00e+00
  COBYLA               0.4963331775      0.00         0       0     0.00e+00
  Differential-Evo     0.4963331775      0.00         0       0     0.00e+00
  Simulated-Ann        0.4963331775      0.00         0       0     0.00e+00

  Pure-state decomposition  [L-BFGS-B]
  ------------------------------------------------------------
     #     weight p_i   C_GME(|psi_i>)      p_i*C_GME
  ------------------------------------------------------
     1   1.0000000000     0.4963331775   0.4963331775
  ------------------------------------------------------
   Sum                                  0.4963331775
  ( best C_GME = 0.4963331775)

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```
================================================================================================================================================
  CLUSTER DETAIL  --  Sites (3, 4, 0, 1, 2)
================================================================================================================================================

  n (body order) = 5
  Purity Tr(r^2) = 1.0000000000
  Rank           = 1
  C_LB (lower)   = 0.0000000000
  C_GME   = 0.4963331775
  C_UB (upper)   = 0.4963331775
  Best C_GME (optimizer)     = 0.4963331775  [L-BFGS-B]
  Bounds : C_LB <= C_GME <= C_UB?     = YES

  Bipartition purities
  ------------------------------------------------------
  A                    B                      Tr(rho_A^2)
  ------------------------------------------------------
  (0,)                 (1, 2, 3, 4)           0.7553574942
  (1,)                 (0, 2, 3, 4)           0.5089904672
  (2,)                 (0, 1, 3, 4)           0.5089904672
  (3,)                 (0, 1, 2, 4)           0.5089904672
  (4,)                 (0, 1, 2, 3)           0.5089904672
  (0, 1)               (2, 3, 4)              0.4335780367
  (0, 2)               (1, 3, 4)              0.4208561822
  (0, 3)               (1, 2, 4)              0.4208561822
  (0, 4)               (1, 2, 3)              0.4208561822
  (1, 2)               (0, 3, 4)              0.4208561822
  (1, 3)               (0, 2, 4)              0.4208561822
  (1, 4)               (0, 2, 3)              0.4208561822
  (2, 3)               (0, 1, 4)              0.3283275934
  (2, 4)               (0, 1, 3)              0.3283275934
  (3, 4)               (0, 1, 2)              0.3283275934

  Method results
  ------------------------------------------------------------------------------
  Method                      C_GME   Time(s)  Restarts   Evals       Spread
  ------------------------------------------------------------------------------
  L-BFGS-B             0.4963331775      0.00         0       0     0.00e+00
  Nelder-Mead          0.4963331775      0.00         0       0     0.00e+00
  Powell               0.4963331775      0.00         0       0     0.00e+00
  COBYLA               0.4963331775      0.00         0       0     0.00e+00
  Differential-Evo     0.4963331775      0.00         0       0     0.00e+00
  Simulated-Ann        0.4963331775      0.00         0       0     0.00e+00

  Pure-state decomposition  [L-BFGS-B]
  ------------------------------------------------------
     #     weight p_i   C_GME(|psi_i>)      p_i*C_GME
  ------------------------------------------------------
     1   1.0000000000     0.4963331775   0.4963331775
  ------------------------------------------------------
   Sum                                  0.4963331775
  ( best C_GME = 0.4963331775)

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================================================================================================================================================
  CLUSTER DETAIL  --  Sites (4, 0, 1, 2, 3)
================================================================================================================================================

  n (body order) = 5
  Purity Tr(r^2) = 1.0000000000
  Rank           = 1
  C_LB (lower)   = 0.0000000000
  C_GME   = 0.4963331775
  C_UB (upper)   = 0.4963331775
  Best C_GME (optimizer)     = 0.4963331775  [L-BFGS-B]
  Bounds : C_LB <= C_GME <= C_UB?     = YES

  Bipartition purities
  ------------------------------------------------------------
  A                    B                       Tr(rho_A^2)
  ------------------------------------------------------------
  (0,)                 (1, 2, 3, 4)            0.5445861398
  (1,)                 (0, 2, 3, 4)            0.7553574942
  (2,)                 (0, 1, 3, 4)            0.7553574942
  (3,)                 (0, 1, 2, 4)            0.7553574942
  (4,)                 (0, 1, 2, 3)            0.7553574942
  (0, 1)               (2, 3, 4)               0.5707360127
  (0, 2)               (1, 3, 4)               0.2795690995
  (0, 3)               (1, 2, 4)               0.2795690995
  (0, 4)               (1, 2, 3)               0.2795690995
  (1, 2)               (0, 3, 4)               0.2795690995
  (1, 3)               (0, 2, 4)               0.2795690995
  (1, 4)               (0, 2, 3)               0.2795690995
  (2, 3)               (0, 1, 4)               0.8768266884
  (2, 4)               (0, 1, 3)               0.8768266884
  (3, 4)               (0, 1, 2)               0.8768266884

  Method results
  ------------------------------------------------------------------------------------
  Method                      C_GME   Time(s)  Restarts   Evals       Spread
  ------------------------------------------------------------------------------
  L-BFGS-B             0.4963331775      0.00         0       0     0.00e+00
  Nelder-Mead          0.4963331775      0.00         0       0     0.00e+00
  Powell               0.4963331775      0.00         0       0     0.00e+00
  COBYLA               0.4963331775      0.00         0       0     0.00e+00
  Differential-Evo     0.4963331775      0.00         0       0     0.00e+00
  Simulated-Ann        0.4963331775      0.00         0       0     0.00e+00

  Pure-state decomposition  [L-BFGS-B]
  ------------------------------------------------------------
     #     weight p_i   C_GME(|psi_i>)      p_i*C_GME
  ------------------------------------------------------
     1   1.0000000000     0.4963331775   0.4963331775
  ------------------------------------------------------
   Sum                                  0.4963331775
  ( best C_GME = 0.4963331775)

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